

Split Complex Lorentzian Dynamics

Part XX: The Unit-Quaternion Core, the Exact Visible Quotient, and the Contraction of the Translation Gap

When a broad spin-like comparison becomes a finite list of explicit isomorphism tests

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Abstract

Part XIX ended at an explanatory boundary rather than a constructive one. The intrinsic reconstruction had produced a visible coincidence quotient, primitive central bridges, half-odd local labels with integer differences, antipodal selector geometry, a minimal global continuity obstruction, and an inherited $2\pi/4\pi$ internal sign structure. Each feature had an established analogue, but the paper deliberately refused to identify the cumulative object with a single classical construction. Task 20 was therefore designed as an equivalence audit: first freeze the intrinsic objects without conventional vocabulary, then permit a comparison layer only after the internal signature is complete.

The result contracts much of the uncertainty. The visible equality relation is proved to be an equivalence relation; its quotient is constructed, the visible map factors through it uniquely, and the quotient is homeomorphic to the actual visible image. That visible image contains the identity and is closed under composition and inversion. Above it, Task 20 constructs two intrinsic internal carriers. The core carrier $\mathcal{L}$ consists exactly of the reference implementers and is closed under product and inversion. The enlarged carrier $\mathcal{L}_Z = \mathcal{C}^\times \mathcal{L}$ is obtained by adjoining all invertible elements of the exact central plane and is again closed. The projection is conjugation. Its kernel on $\mathcal{L}$ is exactly $\{+1, -1\}$, whereas its kernel on $\mathcal{L}_Z$ is the entire invertible centre, which contains a continuous injective one-parameter family. Thus the project contains both a distinguished discrete double ambiguity and a strictly larger continuous central freedom; they are formally different structures and neither may be silently discarded.

The relative-factor problem also closes. Two elements of $\mathcal{L}_Z$ with the same visible projection differ by exactly one invertible central factor, while two elements of the core differ only by sign. The local exact rate set is proved to be a free transitive $\mathbb{Z}$ -set, hence an exact $\mathbb{Z}$ -torsor, and its difference group is precisely $\mathbb{Z}$. Reduction to the full-turn sign loses information: labels separated by an even integer can agree at 2π while producing different central factors at other parameters. The antipodal direction quotient is constructed intrinsically; antipodal-free and antipodally complete domains become fibrewise selector conditions, and unrestricted maximal admissible domains are exactly dense selectors. Selector geometry and the global obstruction share the same reversal involution but are proved not to be the same phenomenon.

Task 20 sharpens the global no-go more dramatically than expected. Half-oddness is not the fundamental source of the contradiction. The smallest intrinsic statement is that a continuous reversal-odd real-valued rate on the connected direction carrier must vanish somewhere. Hence no such rate can take values in any subset of $\mathbb{R}$ that excludes zero; even the nondiscrete positivity condition is impossible. The half-odd lattice supplies local rigidity and the affine label structure, but the global obstruction itself is a continuity-plus-reversal obstruction with a nonvanishing requirement. Equivalently, a quotient-compatible internal choice exists set-theoretically, yet no such choice can be continuous. The global difficulty is therefore no longer an existence problem for representatives; it is the impossibility of a continuous global representative choice over the exact visible quotient.

The single formal external identification of Task 20 is decisive. An explicit injective, unital, multiplicative map from Hamilton's real quaternions into the inherited carrier is constructed,

and its norm-one image is proved to be exactly $\mathcal{L}$. Thus the intrinsic core carrier is *exactly the group of unit quaternions*. Established mathematics identifies the unit quaternions with $SU(2)$ and $Spin(3)$, and their conjugation action on the imaginary quaternion space gives the standard double covering of $SO(3)$ with kernel $\{\pm 1\}$ [2, 5, 14, 3]. The project has not yet formalized the final identification of its visible image with $SO(3)$, so Part XX records this as a near-resolution rather than silently promoting the classical consequence to a Task 20 theorem.

A second near-resolution concerns the enlarged central carrier. The inherited centre is a real two-plane generated by 1 and S with $S^2 = -1$, so its multiplication is complex-like and its invertible part is naturally compared with $\mathbb{C}^\times$. Since the core is now identified with the unit quaternions and the common discrete intersection is $\{\pm 1\}$, the enlarged carrier strongly suggests a central-product description related to $(\mathbb{C}^\times \times SU(2))/\{\pm 1\}$. Separating modulus from phase would place its normalized part near $Spin^c(3) \cong U(2)$ [4, 8, 10, 9]. This identification is not a Task 20 theorem and is therefore reserved as a precise Task 21 test rather than presented as a result.

The broad translation problem of Part XIX has consequently changed character. It is no longer necessary to search simultaneously across descent, covering theory, quaternionic groups, selector geometry, torsors and spin-like behavior without a fixed anchor. The intrinsic core has now been exactly identified; the visible quotient and both kernels have been exactly constructed; the global obstruction has been reduced to continuity of a quotient-compatible choice. The remaining questions are explicit isomorphism and extension questions. Conservation of Difficulty has contracted from “what sort of mathematics is this?” to “which exact classical object does each already-constructed carrier and projection equal?” Task 21 should therefore return to the normal focused mode of the project: one candidate at a time, with explicit maps, kernels, inverses, and negative controls.

Keywords: split-complex Lorentz structure; unit quaternions; $SU(2)$; $Spin(3)$; visible quotient; central extension; double kernel; primitive bridge; $\mathbb{Z}$ -torsor; selector; continuity obstruction; $Spin^c(3)$; $U(2)$; formal verification; Lean 4; conservation of difficulty.

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1 The question inherited from Part XIX

Part XIX was intentionally uncomfortable. The internal development had become precise enough that many conventional comparisons could be stated, but not precise enough to justify collapsing them into one name [11]. The visible coincidence relation resembled a kernel pair. Relation completeness resembled descent. Primitive bridges resembled relative lift factors. The half-odd labels formed an affine $\mathbb{Z}$ -pattern. The reversal quotient was literally a two-orbit quotient on the direction sphere. The global no-go resembled an antipodal continuity obstruction. The $2\pi/4\pi$ internal behavior was spinorially characteristic. Yet no theorem identified the cumulative carrier with a conventional spin group, no standard visible rotation group had been constructed inside the project, and no principal-bundle or characteristic-class problem had been formulated.

The correct Part-XIX conclusion was therefore not that the structure was novel and not that it was “just spin.” It was that the project had reached a *translation gap*: several mature theories appeared to explain different faces of the same reconstruction, while the exact equivalences between those faces were unproved.

Task 20 attacked that gap under a deliberately asymmetric protocol. Phase A was forbidden to use the comparison layer of Task 19 or any conventional target vocabulary. It had to freeze the intrinsic carriers, quotient relations, kernels, rates and local-to-global statements first. Only after that layer was complete could a Phase-B comparison be attempted [1].

This protocol matters for interpreting the outcome. If an established object appears in Phase B, it cannot have shaped the Phase-A definitions. Conversely, an external theorem is not allowed to upgrade an analogy into a project theorem unless an explicit map and the required algebraic/topological properties have actually been proved.

Methodological principle 1 (Part XX reading rule). The strongest conclusion is obtained by keeping three levels separate: *intrinsic Task-20 theorem*, *established classical consequence once an intrinsic object has been identified*, and *next explicit equivalence to be formalized*. Part XX will deliberately state all three, but it will not merge them.

2 Verification boundary and archive provenance

The supplied Aristotle archive reports Task 20 as complete [1]. The run identifier is

2bc3fc02-9b8a-4587-a8bc-41cbd377b212

and the supplied archive has SHA-256

3f8712040e59d8b3d2f4b28af7a8b29bb1e1293b64027b2a6a1182e0e81f5ceb.

The archive records Lean `leanprover/lean4:v4.28.0` and `mathlib v4.28.0` at revision

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The Task 20 audit reports:

- focused build `lake build RequestProject.NullSectorTask20.Task20`: successful;
- whole-project `lake build`: successful, 8249 jobs;
- no warning emitted by a Task 20 module;
- no `sorry`, `admit`, `custom axiom`, `@[implemented_by]`, `native_decide`, `bv_decide`, or `unsafe` in the Task 20 layer;
- all nineteen exposed `task20_*` endpoints report exactly `[propext, Classical.choice, Quot.sound]` under `#print axioms`.

The Task 20 layer contains 2148 lines of Lean source. Its linear source order is

SafeBase $\rightarrow$ LiftCarrier $\rightarrow$ InternalProjection $\rightarrow$ VisibleQuotient $\rightarrow$ BridgeTorsor $\rightarrow$
 ReversalQuotient $\rightarrow$ GlobalObstruction $\rightarrow$ SectionObstruction $\rightarrow$ NegativeControls $\rightarrow$
 Identification $\rightarrow$ Task20.

The firewall is stronger than in the preceding comparison paper. **SafeBase** enters Task 19 through its last reconstruction module only. Task 19’s **Identification** and top-level module are not imported. Task 20’s reconstruction files contain no conventional target words such as **SU(2)**, **SO(3)**, **Spin**, **quaternion**, covering-space terminology, bundle language, cohomology or physical vocabulary. The single comparison module **Identification** is imported only by the top-level endpoint module.

Proof boundary 1 (Build claim versus editorial inspection). *The full 8249-job Lean build is reported by the supplied Aristotle archive. The manuscript environment did not independently rerun the pinned Lean/Lake project. The source order, theorem statements, audit, archive hash, module inventory and prohibited-construct boundary were independently inspected. As before, a green build verifies the encoded statements, not every stronger conventional interpretation suggested by them.*

3 The intrinsic signature is no longer diffuse

Task 20 begins by freezing the objects that had accumulated across the earlier tasks. The central methodological gain is that these objects can now be organized into a small number of levels.

Level	Intrinsic object	Role after Task 20
Visible parameter level	unit direction n and real parameter θ	redundant parametrization of visible transformations
Visible quotient level	$\mathcal{V}$	exact quotient by equality of visible transformations
Internal core level	$\mathcal{L}$	all reference implementers; closed under product/inverse
Extended internal level	$\mathcal{L}_Z = \mathcal{C}^\times \mathcal{L}$	core plus all invertible central factors
Relative level	primitive/relative factors	unique difference between equal visible lifts
Exact local level	rates in $\mathbb{Z} + \frac{1}{2}$	affine labels of locally exact representatives
Reversal level	$n \sim -n$ quotient	selector geometry of direction choices
Global level	continuous quotient-compatible choice	exists set-theoretically, impossible continuously

This table already changes the Part-XIX problem. “Bridge,” “visible relation,” “selector,” “double sign,” and “global no-go” are no longer merely motifs. Each now belongs to an exact carrier and exact map.

4 The visible quotient is constructed, not merely suggested

Let the visible parameter space be the unit-direction carrier crossed with the real parameter. Task 20 defines

$$p \sim q \iff \text{visMap}(p) = \text{visMap}(q), \quad (1)$$

where the visible map is the inherited family $\Phi_n(\theta)$.

The equivalence relation is verified directly and the quotient

$$\mathcal{V} := (\mathbb{S} \times \mathbb{R}) / \sim \quad (2)$$

is constructed.

Theorem 1 (Task 20 visible quotient package). *The visible map factors uniquely through $\mathcal{V}$. The induced map from $\mathcal{V}$ onto the actual visible image is bijective. With the quotient topology on $\mathcal{V}$ and the inherited subspace topology on the visible image, that canonical bijection is a homeomorphism.*

The topological part is not a definitional consequence. The proof uses compactness of the unit-direction space, a new bounded-parameter representation showing that the internal reference elements can be represented with parameter in $[0, 2\pi]$, and the Hausdorff property of the inherited map space. A direct open-map attempt was abandoned; compact-to-Hausdorff homeomorphism supplied the successful route.

Task 20 also proves full closure of the visible image:

$$\text{id} \in G_{\text{vis}}, \quad F, G \in G_{\text{vis}} \Rightarrow F \circ G \in G_{\text{vis}}, \quad (3)$$

with inverses again lying in the visible image. Thus no weakened semigroup or partial-law interpretation is required.

Status 1 (Part XIX near-resolution I). Part XIX asked what the “downstairs” object actually was. Task 20 does not yet assign it a standard group name, but the difficult intrinsic part is closed: the quotient, topology and group closure are constructed. What remains is an explicit isomorphism from this already-fixed visible group to a conventional candidate.

5 Two internal carriers and two different kernels

The internal side becomes equally concrete. Task 20 defines the core carrier

$$\mathcal{L} = \left\{ a \, 1 - J(v) : a^2 + \langle v, v \rangle = 1 \right\}. \quad (4)$$

It then proves the nontrivial representation theorem

$$u \in \mathcal{L} \iff \exists n, \theta : n \text{ unit and } u = U_n(\theta). \quad (5)$$

Hence the apparently coordinate-defined carrier is exactly the set of inherited reference implementers.

Closure under multiplication follows from the reconstructed multiplication law for the J -sector and a proved four-square identity. Closure under inversion is likewise formal. Therefore $\mathcal{L}$ is a genuine group-like internal carrier under the inherited product.

Task 20 then keeps the central freedom rather than normalizing it away. Let $\mathcal{C}^\times$ be the invertible part of the exact central plane generated by 1 and S . The full carrier is

$$\mathcal{L}_Z = \mathcal{C}^\times \mathcal{L}. \quad (6)$$

It too is closed under product and inverse.

The projection is intrinsic conjugation,

$$\text{proj}(u) : x \longmapsto u x u^{-1}, \quad (7)$$

and on the reference family it reproduces the visible transformation:

$$\text{proj}(U_n(\theta)) = \Phi_n(\theta). \quad (8)$$

The kernel theorem is one of the key structural separations of Task 20.

Theorem 2 (Two kernels). *On the core carrier,*

$$\ker(\text{proj}|_{\mathcal{L}}) = \{+1, -1\}. \quad (9)$$

On the full carrier,

$$\ker(\text{proj}|_{\mathcal{L}_Z}) = Z(\mathcal{L}_Z) = \mathcal{C}^\times. \quad (10)$$

Moreover $\{\pm 1\}$ is a proper subset of that centre, and the latter contains a continuous injective one-parameter family.

This immediately corrects a possible overreading of the $2\pi/4\pi$ sign. The project genuinely contains a two-element kernel on a distinguished core, but the total central freedom is larger. There is no theorem saying that the entire internal ambiguity is only $\mathbb{Z}_2$.

The full-turn law remains:

$$U_n(2\pi) = -1, \quad U_n(4\pi) = 1, \quad \text{proj}(U_n(2\pi)) = \text{id}. \quad (11)$$

Thus the discrete kernel is not an artificial subset chosen after the fact; it is visited by the inherited reference one-parameter family itself.

6 Primitive bridges become ordinary relative factors

Task 19 had already made the bridge primitive by removing global exactness from its definition. Task 20 now places that bridge inside the internal projection itself.

Theorem 3 (Unique relative central factor). *Let $u, u' \in \mathcal{L}_Z$ and suppose*

$$\text{proj}(u) = \text{proj}(u'). \quad (12)$$

Then there exists exactly one $z \in \mathcal{C}^\times$ such that

$$u = z u'. \quad (13)$$

If $u, u' \in \mathcal{L}$, the relative factor reduces to a sign:

$$u = u' \quad \text{or} \quad u = -u'. \quad (14)$$

No continuity and no exactness hypothesis is used.

This theorem clarifies what the bridge is before any conventional comparison is attempted. It is not an extra object attached to local families by hand. It is the unique element of the internal kernel relating two lifts of the same visible datum. The larger carrier remembers the full central factor; the core remembers only the sign.

Exactness then selects a narrower subset of those relative factors. The Task-19 normal form remains unchanged:

$$c(n, \theta) = z_{\alpha(n), \beta(n)}(\theta). \quad (15)$$

Local exactness is equivalent to

$$\alpha(n) = 0, \quad \beta(n) \in \mathbb{Z} + \frac{1}{2}, \quad (16)$$

while global exactness adds reversal oddness. Thus the vanishing of α and the discretization of β are not properties of the internal kernel itself; they are properties imposed by exact representative independence.

7 The half-odd labels are an exact torsor, not a disguised sign

Task 20 formalizes the local exact rate set as

$$H := \mathbb{Z} + \frac{1}{2}. \quad (17)$$

The integers act by translation.

Theorem 4 (Exact affine structure). *The $\mathbb{Z}$ -action on H is free and transitive. For any $x, y \in H$ there exists exactly one $k \in \mathbb{Z}$ such that*

$$x + k = y. \quad (18)$$

The set of all differences $y - x$ is exactly $\mathbb{Z}$. No element of H is intrinsically distinguished as an origin.

This closes one of the comparison questions from Part XIX at the purely algebraic level. The phrase “torsor-like” can be replaced by the exact statement: the allowed local exact labels are a torsor over $\mathbb{Z}$.

Task 20 then proves that reduction to a discrete sign is insufficient to reconstruct the full central data. The labels $1/2$ and $5/2$ differ by an even integer. Their full-turn central values agree, but at $\theta = \pi/2$ their central factors differ. More generally, distinct labels give distinct one-parameter factors.

Methodological principle 2 (The sign is a quotient of information, not the whole information). The two-element kernel of the core and the integer/half-integer label data coexist but are not interchangeable. Full-turn evaluation collapses infinitely many distinct central one-parameter factors to the same sign. Any later reduction modulo 2 must therefore be justified by the specific observable or quotient being studied; it cannot be treated as lossless.

8 The reversal quotient and maximal selectors

Direction reversal is treated in Task 20 as an involution on the unit-direction carrier. Its orbit relation is constructed without importing a conventional name, giving a quotient $\mathcal{R}$.

For a unit-direction set D , Task 20 proves the exact fibre dictionary:

$$D \text{ antipodal-free} \iff \text{at most one point of } D \text{ in each fibre}, \quad (19)$$

$$D \text{ antipodally complete} \iff \text{at least one point of } D \text{ in each fibre}, \quad (20)$$

$$D \text{ both} \iff \text{exactly one point of } D \text{ in each fibre}. \quad (21)$$

Thus the selector language that was only explanatory in Part XIX is now exact internally. The unusual Task-19 maximality theorem becomes

$$D \text{ maximal admissible} \iff D \text{ is a dense selector of } \mathcal{R}. \quad (22)$$

Task 20 separates the two ingredients. The selector property comes solely from the involution quotient. Density does not. Density enters because a continuous half-odd ambient rate on a dense set becomes globally rigid. Therefore maximal admissibility is a hybrid theorem combining quotient geometry with rate extension.

Maximal selectors remain nonunique. Task 20 preserves an explicit pair related by reversal and identifies a fibre on which the two choices differ.

Status 2 (Part XIX near-resolution II). The formerly “foreign” dense-selector theorem is no longer mysterious at the level of its ingredients. Its selector half is pure orbit geometry; its density half is rate rigidity. What remains open is whether this particular hybrid class has a standard external classification useful enough to replace the intrinsic theorem.

9 The global no-go is simpler than the half-odd lattice

The most important conceptual simplification of Task 20 is the sharpening of the global obstruction. Parts XVIII and XIX had isolated the incompatible triple

$$\text{continuity} + \text{half-oddness} + \text{reversal oddness}. \quad (23)$$

Task 19 proved that this triple was minimally unsatisfiable. Task 20 asks which property of half-oddness is actually used.

The answer is: only exclusion of zero.

Theorem 5 (Minimal intrinsic zero theorem). *Let $f : \mathbb{S} \rightarrow \mathbb{R}$ be continuous and reversal-odd:*

$$f(-n) = -f(n). \quad (24)$$

Then there exists $n_0 \in \mathbb{S}$ such that

$$f(n_0) = 0. \quad (25)$$

Consequently, if $A \subset \mathbb{R}$ excludes zero, there is no continuous reversal-odd rate whose values all lie in A .

The proof uses connectedness/preconnectedness and the intermediate value property. No discreteness is needed. Task 20 includes the nondiscrete negative control

$$\nexists f : \mathbb{S} \rightarrow (0, \infty) \quad \text{continuous and reversal-odd}. \quad (26)$$

This materially changes the interpretation of the half-odd lattice.

Status 3 (What half-oddness does and does not do). Half-oddness is essential for the local label torsor, integer relative transitions, local constancy, and exact one-parameter classification. It is *not* the fundamental source of the global nonexistence. The global obstruction already exists for any nonvanishing target condition. The half-odd lattice is a refinement of the local structure, not the primitive topological obstruction.

This result also narrows the external comparison. The global contradiction no longer requires a special theorem about a discrete affine lattice. It belongs to the elementary family of odd-map zero theorems. More sophisticated antipodal topology may repackage the result, but it is not required to prove it.

10 The sharpest formulation: choice exists, continuous choice does not

Task 20 then packages the entire global difficulty in a way that removes the remaining ambiguity between existence, exactness and continuity.

A quotient-compatible internal choice is a map σ from the visible parameter carrier into the core $\mathcal{L}$ such that:

1. $\sigma(p)$ lies in the internal core;
2. $\text{proj}(\sigma(p))$ is the visible transformation represented by p ;
3. if two parameters have the same visible image, they receive exactly the same internal value.

Task 20 proves both halves:

$$\exists \sigma \text{ quotient-compatible,} \quad (27)$$

but

$$\nexists \sigma \text{ quotient-compatible and continuous.} \quad (28)$$

The proof of noncontinuity uses only the sign ambiguity, connectedness of the parameter carrier, and the inherited full-turn identity

$$U_n(2\pi) = -1 \neq 1 = U_n(0) \quad (29)$$

while both project to the same visible identity.

Methodological principle 3 (The global phenomenon after Task XX). The obstruction is no longer correctly described as “no global lift” or “no global representative.” A global set-theoretic representative choice exists. What fails is the existence of a *continuous* quotient-compatible representative choice.

This is the point at which the bridge, rate and relation-completeness languages truly collapse into one global problem. Task 19 had already proved

$$\begin{aligned} &\text{global exact primitive bridge} \\ &\iff \text{global exact jointly regular family} \\ &\iff \text{relation-complete locally exact open system.} \end{aligned} \quad (30)$$

Task 20 adds the quotient language: all are presentations of the impossibility of making the internal choice continuous while respecting the visible quotient.

11 The formal external breakthrough: the core is the unit-quaternion group

Phase B of Task 20 permits an external comparison only after the intrinsic signature is frozen. Aristotle makes exactly one such comparison formal.

Let $\mathbb{H}$ denote Hamilton’s real quaternion algebra. Define

$$\text{fromQuat}(q) = q_{\text{re}} 1 - J(q_i, q_j, q_k). \quad (31)$$

Task 20 proves:

$$\text{fromQuat}(1) = 1, \quad (32)$$

$$\text{fromQuat}(qp) = \text{fromQuat}(q) \text{fromQuat}(p), \quad (33)$$

$$\text{fromQuat} \text{ is injective,} \quad (34)$$

and, most importantly,

$$\mathcal{L} = \text{fromQuat}(\{q \in \mathbb{H} : \text{normSq}(q) = 1\}). \quad (35)$$

Theorem 6 (Task 20 exact external identification). *The intrinsic core carrier $\mathcal{L}$, with its inherited product, is exactly the group of unit quaternions under the explicit map (31).*

This theorem resolves the most important open comparison from Part XIX. The core is not merely quaternion-like: the multiplication law, carrier and unit sphere coincide under an explicit injective map.

The provenance is equally important. The quaternion library is imported only in the comparison module. The four-square norm identity needed to prove closure of the intrinsic core had already been proved inside the reconstruction coordinates. Quaternion terminology is therefore a recognition of an already closed multiplication law, not an input used to build it.

12 What established mathematics now says immediately – and what the project has not yet formalized

Once the exact unit-quaternion identification is available, several established facts become relevant in a much stronger way than they were in Part XIX. The group of unit quaternions is classically isomorphic to $SU(2)$ and to $Spin(3)$. Conjugation by a unit quaternion on the three-dimensional space of imaginary quaternions gives a surjective group homomorphism onto $SO(3)$ with kernel $\{\pm 1\}$; equivalently,

$$SU(2)/\{\pm I\} \cong SO(3), \quad (36)$$

and $SU(2)$ is the standard double cover of $SO(3)$ [2, 5, 14, 3].

This established theorem matches an unusually large portion of the already-proved intrinsic signature:

Task-20 intrinsic fact	Classical unit-quaternion fact
$\mathcal{L}$ is unit quaternions	$S^3 \cong SU(2) \cong Spin(3)$
projection is conjugation	rotations arise by quaternion conjugation
core kernel is $\{\pm 1\}$	kernel of $SU(2) \rightarrow SO(3)$ is $\{\pm I\}$
2π gives -1 , 4π gives $+1$	standard double-cover full-turn behavior
visible image is a group	target of standard conjugation is $SO(3)$
no continuous quotient-compatible choice	nontrivial double cover has no global continuous section

The comparison is now much more than a thematic resemblance. However, only the first row is an explicit Task 20 external identification. The project has not yet constructed a conventional $SO(3)$ object and proved that the Task-20 visible image is isomorphic to it; nor has it proved that its intrinsic projection is exactly the standard $Spin(3) \rightarrow SO(3)$ map under both identifications.

Proof boundary 2 (Near-resolution, not yet project theorem). *From established mathematics, “unit quaternions” already carries the names $SU(2)$ and $Spin(3)$. It is therefore reasonable in comparison prose to say that Task 20 has identified the core with the classical group underlying $Spin(3)$. What remains unproved inside the project is the commuting identification of the entire projection: visible image $\cong SO(3)$ and intrinsic $proj =$ standard double-cover projection under the chosen isomorphisms.*

This distinction is exactly why Task 21 can now be narrow. It no longer needs to ask “is the object spin-like?” It can ask for one commutative diagram and either prove or refute it.

13 A second near-resolution: the enlarged central carrier

The full carrier $\mathcal{L}_Z$ contains structure not present in the core double cover. Its kernel is the invertible central plane $\mathcal{C}^\times$, not merely $\{\pm 1\}$.

The inherited central multiplication already has the form

$$(a1 + bS)(c1 + dS) = (ac - bd)1 + (ad + bc)S, \quad S^2 = -1. \quad (37)$$

At the ordinary algebraic level, this is the multiplication law of the complex plane. The invertible elements therefore invite the comparison

$$\mathcal{C}^\times \leftrightarrow \mathbb{C}^\times. \quad (38)$$

This comparison has not been packaged as a Task 20 formal external theorem, but unlike the broad analogies of Part XIX it now requires only an explicit two-coordinate algebra isomorphism.

Because $\mathcal{C}^\times$ is central, $\mathcal{L}$ is the unit-quaternion core, and their discrete intersection contains the common signs $\{\pm 1\}$, the natural next candidate for the full carrier is a central product of the form

$$(\mathbb{C}^\times \times SU(2))/\{(1, 1), (-1, -1)\}. \quad (39)$$

Separating a nonzero complex number into positive modulus and unit phase gives

$$\mathbb{C}^\times \cong \mathbb{R}_{>0} \times U(1). \quad (40)$$

The normalized central product then resembles

$$Spin^c(3) = (Spin(3) \times U(1))/\mathbb{Z}_2, \quad (41)$$

which is classically isomorphic to $U(2)$ [4, 8, 10, 9]. This suggests the provisional comparison

$$\mathcal{L}_Z \stackrel{?}{\cong} \mathbb{R}_{>0} \times Spin^c(3) \cong \mathbb{R}_{>0} \times U(2). \quad (42)$$

Methodological caveat 1 (Task-21 candidate only). *Equation (42) is not a Task 20 theorem. Task 20 proves the ingredients that make the candidate precise: a unit-quaternion core, a central invertible two-plane, a full carrier generated by their products, a continuous central kernel, and a discrete $\{\pm 1\}$ core kernel. It does not prove the exact intersection, quotient universal property, topology, or explicit isomorphism required for (42). These are exactly the missing obligations for Task 21.*

This candidate is nevertheless valuable because it explains why the Part-XIX phrase “one single classical object” was too coarse. There may be a classical core double cover and a larger classical central extension simultaneously. The project should not expect one kernel to describe both.

14 Three Part-XIX near-identifications revisited

Part XIX deliberately used a strict three-column comparison: what had been reconstructed, what established object it resembled, and whether the comparison was exact. Task 20 changes several entries of that table. It is useful to state the changes directly because they explain why the project can leave the broad survey mode.

Part-XIX question	Task-20 resolution	Residual obligation
Is visible equality merely an informal equivalence relation?	No. The exact quotient is constructed and is homeomorphic to the visible image.	Give that already-fixed visible group a standard external name by explicit isomorphism.
Is the internal carrier only spinorially reminiscent?	No at the core level. The core is exactly the group of unit quaternions.	Formalize the standard $SU(2)/Spin(3)$ model and compare the projection, not merely the carrier.
Is the central bridge merely analogous to a lift factor?	No intrinsically. Equal visible lifts differ by a unique element of the internal kernel.	Identify the full extension to which the kernel belongs.
Is the half-integer/integer pair only torsor-like?	No. It is exactly a free transitive $\mathbb{Z}$ -set with difference group $\mathbb{Z}$.	Determine its representation-theoretic meaning inside the identified groups.

Part-XIX question	Task-20 resolution	Residual obligation
Is the global no-go caused by half-integer discreteness?	No. Any continuous reversal-odd real rate has a zero.	Translate the quotient-compatible-choice theorem to a standard section theorem after the group identification.
Are maximal admissible sets mysterious selectors?	The selector statement is exact; density comes from rate rigidity.	Decide whether the hybrid “dense selector + extension constraint” has a useful standard classification.
Is one classical object already enough?	Not proved; core and full carrier have different kernels.	Test the core double cover and enlarged central extension separately.

The table shows that Part XIX did not fail to recognize an obvious answer. It stopped one theorem before recognition was safe. Task 20 supplies that theorem for the core and enough structure to make the remaining identifications local rather than speculative.

15 Why the quaternion identification is not circular

Because the unit-quaternion result is so suggestive, its provenance deserves its own audit. A conventional derivation could begin with the quaternion algebra, restrict to norm-one elements, use the standard conjugation action, and immediately obtain the familiar double-valued rotation picture. That is not the order used here.

Before the comparison module is imported, the reconstruction already contains:

1. the three-dimensional generator map J ;
2. the multiplication identity

$$J(n)J(m) = -\langle n, m \rangle 1 - J(n \times m); \quad (43)$$

3. the reference one-parameter family

$$U_n(\theta) = \cos(\theta/2) 1 - \sin(\theta/2)J(n); \quad (44)$$

4. closure of the intrinsic carrier $a1 - J(v)$ with $a^2 + \langle v, v \rangle = 1$ under the inherited product;
5. a four-square norm identity proved in the intrinsic coordinates;
6. the conjugation projection and its core kernel $\{\pm 1\}$.

Only after these statements build does the comparison module import `Mathlib.Algebra.Quaternion` and define `fromQuat`. The external quaternion coordinates then match the already reconstructed scalar/vector split, and multiplicativity follows by comparing the two multiplication formulas. In other words, the quaternion algebra is a *recognition theorem* for an existing carrier.

Methodological principle 4 (Recognition versus construction). A familiar object reconstructed without its name is not automatically new, but identifying it later is scientifically different from having assumed it at the start. The recognition records which earlier constraints were sufficient to force that familiar multiplication law.

This distinction is central to the larger project. If Task 21 succeeds in identifying the intrinsic projection with the standard $Spin(3) \rightarrow SO(3)$ map, the result will not say that the project rediscovered the existence of that cover. It will say that the earlier implementation and coherence constraints were already strong enough to force a carrier and projection equivalent to it.

16 Two different two-valued structures must remain separate

The cumulative development now contains at least two natural two-valued phenomena:

1. the internal kernel of the core projection,

$$\ker(\text{proj}|\mathcal{L}) = \{\pm 1\}; \quad (45)$$

2. the reversal involution on directions,

$$n \longmapsto -n, \quad (46)$$

whose fibres are the two-element direction pairs used in the selector theory.

Both are copies of a two-element action in a loose sense, but Task 20 does not identify them as the same action. They live on different carriers: the first is a kernel in the internal group, the second is an involution on the direction sphere. Their mathematical roles are also different. The first measures the ambiguity of an internal implementer after visible projection; the second forgets orientation of a spatial axis.

They are nevertheless coupled by the inherited axis-angle parametrization. Reversing the axis and changing the sign/shift of the parameter can represent the same visible transformation, and the exact coincidence theorem is what transports that relation into the internal sign ambiguity. Task 19 then shows that reversal-related visible coincidences are the only nonautomatic relation class after same-direction and identity-degenerate relations are discharged.

This provides a precise reformulation of one of the “foreign connections” from Part XIX:

Two-valued object	What Task 20 actually proves
core sign $\{\pm 1\}$	exact kernel of internal projection
direction pair $\{n, -n\}$	exact fibre of reversal quotient
relation between them	mediated by visible coincidence/axis-angle laws, not identified as equality of actions
selector obstruction	uses direction involution plus topology/rate constraints
continuous-lift obstruction	uses core sign plus quotient compatibility and full-turn path

If Task 21 identifies the core projection with the standard quaternionic double cover, it should still separately prove how the axis-reversal quotient sits inside the classical axis-angle parametrization of the target group. This is exactly the kind of sharp residual question that was unavailable in Part XIX.

17 Core double cover versus enlarged central extension

Another Part-XIX ambiguity is now resolved negatively: there is no reason to expect the same kernel to describe the core and the full internal carrier. Task 20 proves that it does not.

The core sequence has the intrinsic shape

$$\{\pm 1\} \longrightarrow \mathcal{L} \xrightarrow{\text{proj}} G_{\text{vis}}, \quad (47)$$

whereas the full sequence has the shape

$$\mathcal{C}^\times \longrightarrow \mathcal{L}_Z \xrightarrow{\text{proj}} G_{\text{vis}}. \quad (48)$$

The first kernel is discrete; the second is continuous. The core is contained in the full carrier, but the two extensions encode different levels of representative freedom.

This suggests a useful organizational principle for all later comparisons:

1. identify the *core cover* first;
2. identify the *central enlargement* second;
3. only then ask which local labels/bridges correspond to one-parameter subgroups of the enlargement.

Attempting to identify the full carrier directly with a simple double cover would erase the continuous kernel and contradict the formal Task-20 theorem. Conversely, using only the full continuous kernel would hide the distinguished discrete core sign and the $2\pi/4\pi$ reference law. The data demand a nested description.

This nested picture is one reason the $Spin^c(3)$ candidate in section 13 is attractive as a *test*: classical $Spin^c$ groups are themselves central enlargements of spin groups by a circle factor modulo the common sign. The additional positive-modulus direction suggested by $\mathbb{C}^\times$ would then account for why the Task-20 full centre is larger than a normalized $U(1)$ kernel. But only an explicit Task-21 quotient theorem can decide whether this analogy is exact.

18 Why Part XIX was almost resolved, but correctly stopped short

Part XIX ended with the statement that the reconstructed structure had entered the same mathematical ecosystem as several established theories without yet being identified with one. Task 20 shows that this statement was conservative in exactly the right way.

Three broad Part-XIX uncertainties have contracted:

1. **What is the internal core?** Closed: it is exactly the unit-quaternion group.
2. **What is the visible object intrinsically?** Closed up to external naming: it is the exact quotient by visible equality, homeomorphic to the visible image, and that image has full group closure.
3. **What is the global obstruction really about?** Closed: set-theoretic quotient-compatible choice exists; continuity is what fails.

Thus the explanatory problem no longer has to float among many analogies. The remaining translation questions have explicit source and target objects.

At the same time, Task 20's own Phase-B verdict — “no single established object identifies the complete reconstructed structure at this standard of proof” — must be read literally. The comparison module formalized only the unit-quaternion identification. It deliberately did not construct a rotation group, covering map, principal bundle or cohomological object. The absence of those formal identifications is a theorem-boundary statement, not evidence that such identifications are false.

Likewise the audit statement that no matching combined formulation was located in the searched literature is not a novelty theorem. The archive does not provide a comprehensive negative literature-search log comparable to the formal proof ledger. Established low-dimensional group theory already supplies the chain

$$\text{unit quaternions} \cong SU(2) \cong Spin(3) \longrightarrow SO(3), \quad (49)$$

with two-element kernel, and established $Spin^c$ theory supplies

$$Spin^c(3) \cong (SU(2) \times U(1))/\mathbb{Z}_2 \cong U(2) \quad (50)$$

[5, 6, 4, 8, 10].

These facts do not finish the project comparison. They do, however, mean that the next work should be a formal identification audit against these explicit candidates, not another open-ended search for a vocabulary.

Methodological principle 5 (Why normal targeted excavation can resume). Part XIX needed a broad equivalence audit because the anchors were missing. Task 20 supplies the anchors. From now on, each comparison can again be posed as a narrow theorem: construct an explicit map, prove multiplicativity/continuity, compute its kernel and image, and test whether the required diagram commutes. Failure is as informative as success.

19 The visible quotient, relation completeness, and the classical section question

The quotient-compatible-choice theorem also clarifies the relation-completeness branch. In Part XIX, relation completeness was naturally compared with descent but had not been reduced to a conventional quotient section statement. Task 20 now gives the exact quotient $\mathcal{V}$ and the exact internal core $\mathcal{L}$.

If Task 21 identifies the visible group with $SO(3)$ and the core projection with the standard double cover, then the Task-20 theorem

$$\exists \text{ quotient-compatible choice,} \quad \nexists \text{ continuous quotient-compatible choice} \quad (51)$$

would become the familiar distinction between a set-theoretic section and a continuous section of a nontrivial double covering.

But the project should prove this translation rather than import it. In particular, Task 20's σ is formulated on the redundant parameter carrier and required to be constant on visible-equality classes. To turn it into a literal section

$$s : G_{\text{vis}} \longrightarrow \mathcal{L} \quad (52)$$

of the intrinsic projection, one must prove the factorization through $\mathcal{V}$ and then through the homeomorphism with the visible image. The intrinsic ingredients for this are now available.

This is one of the clearest examples of a Part-XIX “almost resolution.” The descent language and the covering-section language are no longer two vague analogies. Task 21 can test whether they are exactly the same theorem after the visible/core group identifications.

20 What remains genuinely independent after the contraction

Task 20 does not collapse every branch of Part XIX. Three residual questions remain structurally separate unless later theorems connect them.

20.1 Disconnected label extension

The Task-19 closure-extension criterion is merely frozen and re-exported. Prescribed half-odd labels on a disconnected domain are realizable exactly when they extend continuously to the closure with the required reversal relation. Pairwise separation of differently labelled closures remains necessary but not sufficient.

Task 20 does not prove that this extension problem is independent of the quotient/bridge problem, nor does it prove that they are the same. Their hypotheses are disjoint after local rate classification, but that is not an independence theorem.

20.2 Analytic regularity

The complete smooth ladder remains

$$C^0 = C^1 = \dots = C^k = C^\infty \quad (53)$$

at the level of arbitrary-domain admissibility. Analytic plateau rigidity still blocks the existing staircase proof, while antipodal-free domains retain analytic representatives. Arbitrary-domain analytic admissibility remains open.

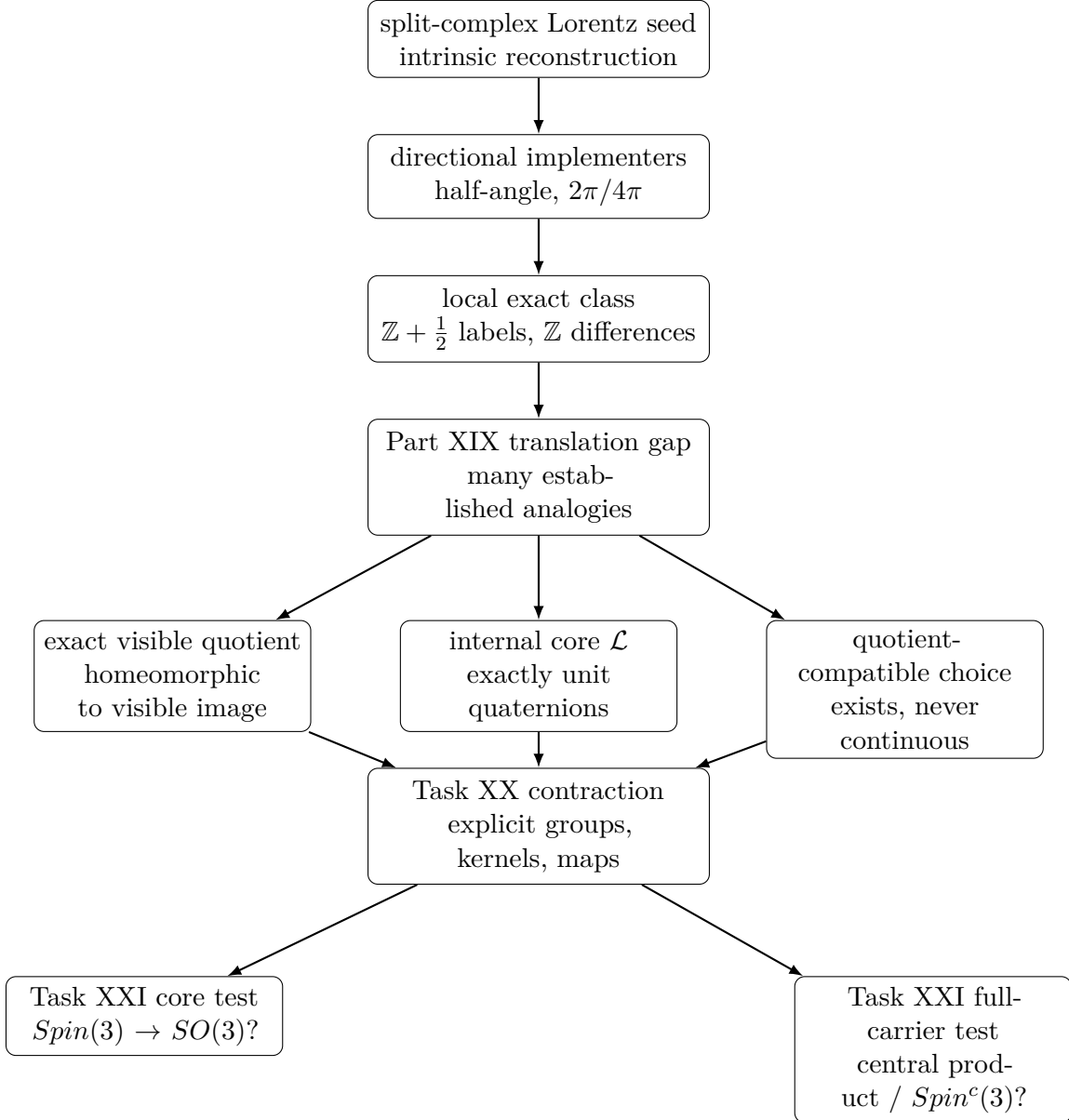
20.3 Maximal preconnected geometry

The unrestricted maximal admissible class is exactly the dense selectors. The maximal preconnected/open-preconnected geometry, however, was already only partially classified in Task 19 and Task 20 does not close it. The core group identification does not automatically classify those point-set domains.

These residual branches are important because they prevent an overcompressed conclusion. Even if Task 21 identifies the core projection with the standard $Spin(3) \rightarrow SO(3)$ double cover, it would not follow that every domain/extension theorem is merely a standard spin theorem. Some results involve the additional ambient rate and selector constraints of this reconstruction.

21 The new Conservation-of-Difficulty geometry

The history from Part XVI to Part XX can now be drawn differently. What looked in Part XIX like an expanding network of partially matching established ideas has begun to contract around a concrete group-theoretic core.



The key conservation statement is no longer that every solved problem creates a new obstruction

of the same kind. Task 20 solves part of the explanatory problem by identifying the core. Difficulty is conserved because the identification exposes a finite list of exact maps that must either commute or fail.

Methodological principle 6 (Contracted Conservation of Difficulty). Once a formerly diffuse comparison acquires an exact intrinsic carrier and an exact classical identification of that carrier, the next difficulty is not conceptual vocabulary. It is diagram chasing in the literal sense: identify the target group, compute kernels and intersections, prove quotient properties, and determine which extra central or topological data survive beyond the core identification.

This is why the project can “go back to normal.” The next tasks can again be deliberately narrow and falsifiable.

22 Claim boundaries and warnings

Part XX contains stronger conventional discussion than Part XIX, so the boundaries must be unusually explicit.

22.1 Unit quaternions are formally identified; the name $Spin(3)$ is classical comparison

The Lean theorem identifies $\mathcal{L}$ with the norm-one quaternions. The isomorphisms unit quaternions $\cong SU(2) \cong Spin(3)$ are established external mathematics. They are not independently encoded as Task 20 Lean structures. A paper may cite the classical identification, but it must not report a Task-20 theorem named “Lift = Spin(3)” until the corresponding formal object/map is built.

22.2 The visible image is not yet formally $SO(3)$

The visible image is a topological group under inherited composition and is the image of quaternionic conjugation after the core identification. This is strongly aligned with the standard $SO(3)$ construction, but an explicit project-level isomorphism to a conventional $SO(3)$ carrier is still missing.

22.3 The full carrier is not yet $Spin^c(3)$ or $\mathbb{R}_{>0} \times U(2)$

The candidate in section 13 is precise enough to test, not precise enough to claim. The project must prove an explicit complex-plane isomorphism for $\mathcal{C}^\times$, determine $\mathcal{C}^\times \cap \mathcal{L}$, construct the central-product quotient map, and verify the topology and multiplication before using the conventional group name.

22.4 No spin structure has been constructed

Even a successful identification of the core with $Spin(3)$ is a group-level statement. A classical spin *structure* is a lift of an oriented orthonormal frame bundle. The project has not reconstructed such a principal bundle or a second Stiefel–Whitney obstruction. The word “spin” must therefore continue to distinguish group identification from spin-structure classification [6, 7].

22.5 The global no-go is not fundamentally half-integer or cohomological

Task 20 proves that the primitive no-go needs only continuity, reversal oddness and exclusion of zero. Any later representation-theoretic or cohomological interpretation of the half-odd labels must preserve this fact rather than retroactively making the lattice the source of the global obstruction.

23 Status table after Task XX

Question	Part-XX status
Visible equality is an equivalence and quotient exists?	PROVED.
Quotient equals visible image topologically?	PROVED: canonical homeomorphism.
Visible image closed under identity/composition/inversion?	PROVED.
Internal core closed and equal to reference implementers?	PROVED.
Full central carrier closed?	PROVED.
Core kernel?	Exactly $\{\pm 1\}$, PROVED.
Full kernel?	Entire invertible centre, PROVED; continuous.
Relative factor between equal visible lifts?	Unique central factor; sign in core, PROVED.
Local exact labels a genuine $\mathbb{Z}$ -torsor?	PROVED.
Does mod-2/full-turn sign retain the full label data?	No, explicitly REFUTED.
Reversal quotient selector dictionary?	PROVED.
Maximal unrestricted admissible domains?	Exactly dense selectors, PROVED.
Primitive global obstruction needs half-odd discreteness?	No, REFUTED; zero-exclusion suffices.
Set-theoretic quotient-compatible internal choice?	EXISTS, PROVED.
Continuous quotient-compatible internal choice?	DOES NOT EXIST, PROVED.
Relation-complete system / exact bridge / exact global family?	Equivalent inherited formulations, PROVED.
Internal core identified externally?	EXACTLY: unit quaternions.
Internal core formally identified with $Spin(3)$?	NOT YET as a project theorem; classical consequence of unit-quaternion identification.
Visible image formally identified with $SO(3)$?	OPEN / targeted next test.
Intrinsic projection formally identified with standard $Spin(3) \rightarrow SO(3)$?	OPEN / targeted next test.
Full carrier identified with a standard central extension?	PARTIAL only; targeted next test.
$Spin^c(3) \cong U(2)$ candidate relevant?	Classical candidate, not a Task-20 theorem.
Disconnected label extension geometric classification?	OPEN.
Arbitrary-domain analytic admissibility?	OPEN.
Spin structure or w_2 obstruction constructed?	NO.

24 Task XXI: return to focused equivalence proofs

Task 21 should be narrower than Task 20. The broad audit has done its job. The next task should test the explicit classical candidates exposed by the exact unit-quaternion identification without assuming that all tests succeed.

A suitable neutral title is:

Task XXI – Core and Central-Extension Identification: The Quaternionic Projection, the Visible Rotation Group, and the Full Central Carrier.

The task should retain a two-layer firewall: intrinsic objects imported unchanged, conventional groups introduced only in an identification module or explicitly named comparison package. The following work packages are the natural focused sequence.

24.1 WP1: formalize the core as a standard quaternionic Lie group

Package the Task-20 map from unit quaternions to $\mathcal{L}$ as an explicit multiplicative equivalence, including inverse and topology if the available library permits it. Then compare the unit-quaternion group with one standard named model, preferably $SU(2)$ or $Spin(3)$, using an explicit existing equivalence rather than only literature prose.

Required verdict:

$$\mathcal{L} \cong SU(2) ? \quad \text{and/or} \quad \mathcal{L} \cong Spin(3) ?$$

with exact status ‘PROVED’, ‘PARTIAL’, or ‘NOT FORMALIZED’.

24.2 WP2: identify the visible group

Construct a conventional $SO(3)$ candidate in the comparison layer and define the map from the intrinsic visible image to it by restricting the already reconstructed action to the three-dimensional J -sector (or another intrinsic three-dimensional invariant subspace if that is the correct carrier).

Prove or refute:

$$G_{\text{vis}} \cong SO(3). \quad (54)$$

The proof must establish the preserved quadratic form, orientation/determinant condition, surjectivity onto the candidate, and an explicit inverse or sufficient classification theorem.

24.3 WP3: identify the core projection

Under the WP1 and WP2 equivalences, test whether the diagram

$$\begin{array}{ccc} \mathcal{L} & \xrightarrow{\text{proj}} & G_{\text{vis}} \\ \downarrow & & \downarrow \\ Spin(3) \text{ or } SU(2) & \longrightarrow & SO(3) \end{array} \quad (55)$$

commutes, where the lower map is the standard quaternionic conjugation/double-cover map.

The existing intrinsic kernel theorem $\ker = \{\pm 1\}$ must be recovered as the kernel of the lower map rather than used to assume the identification.

24.4 WP4: turn the continuity no-go into a section theorem

Using the homeomorphism $\mathcal{V} \cong G_{\text{vis}}$, factor Task 20’s quotient-compatible choice through the visible quotient. Define the corresponding set-theoretic section of the core projection and prove that no continuous section exists.

If WP3 succeeds, compare this theorem exactly with the standard no-global-continuous-section statement for the resulting nontrivial double cover. Determine whether Task 19 relation completeness is then literally an effective section/descent condition or retains extra hypotheses.

24.5 WP5: identify the exact central plane

Construct explicitly

$$a \, 1 + bS \longleftrightarrow a + ib \quad (56)$$

and prove the algebraic and topological properties needed to identify the invertible central carrier with $\mathbb{C}^\times$.

Do not infer the result solely from $S^2 = -1$; prove injectivity, surjectivity onto the central plane and multiplicativity using the inherited coefficient-uniqueness theorem.

24.6 WP6: classify the full carrier as a central product

Determine exactly

$$\mathcal{C}^\times \cap \mathcal{L}. \quad (57)$$

Test whether it is precisely $\{\pm 1\}$. If so, construct the multiplication map

$$\mathbb{C}^\times \times \mathcal{L} \longrightarrow \mathcal{L}_Z \quad (58)$$

and compute its kernel. Prove or refute the quotient description

$$\mathcal{L}_Z \cong (\mathbb{C}^\times \times \mathcal{L}) / \mathbb{Z}_2. \quad (59)$$

24.7 WP7: test the $Spin^c(3) / U(2)$ candidate

Only after WP1, WP5 and WP6 succeed, split $\mathbb{C}^\times$ into positive modulus and unit phase and test whether the normalized quotient is exactly

$$(Spin(3) \times U(1)) / \mathbb{Z}_2 = Spin^c(3) \cong U(2). \quad (60)$$

Then determine whether the full carrier is isomorphic to $\mathbb{R}_{>0} \times Spin^c(3)$ or whether the inherited multiplication/topology differs in a specific way.

A negative result is fully acceptable and should identify the first failed group law, kernel or topology condition.

24.8 WP8: locate the torsor data inside the identified groups

If WP3 or WP7 succeeds, classify the half-odd local labels and integer differences relative to one-parameter subgroups/characters/logarithmic lifts of the identified group. Test which information is retained under the central sign quotient and explain formally the Task-20 theorem that mod-2 reduction loses information.

Do not identify the $\mathbb{Z}$ -torsor with $H^1(-; \mathbb{Z}_2)$ or any spin-structure classification unless an actual bundle/cohomology object has been constructed.

24.9 WP9: preserve independent branches

Do not force disconnected extension or analytic admissibility into the group-identification package. Revisit them only if one of WP1–WP8 supplies a theorem that genuinely constrains them.

24.10 Task-XXI decision table

The final Task 21 report should give explicit verdicts for at least:

Question	Required verdict
$\mathcal{L}$ formally equivalent to $SU(2)$?	PROVED / PARTIAL / NOT FORMALIZED
$\mathcal{L}$ formally equivalent to $Spin(3)$?	PROVED / PARTIAL / NOT FORMALIZED
Visible image equivalent to $SO(3)$?	PROVED / REFUTED / PARTIAL
Intrinsic projection equals standard double-cover map?	PROVED / REFUTED / PARTIAL
Kernel correspondence $\{\pm 1\}$ exact under the identification?	PROVED / REFUTED
Task-20 internal choice becomes a section of the identified projection?	PROVED / PARTIAL

Question	Required verdict
No-continuous-choice theorem equals no-continuous-section theorem?	PROVED / PARTIAL / REFUTED
$\mathcal{C}^\times \cong \mathbb{C}^\times$?	PROVED / REFUTED
$\mathcal{C}^\times \cap \mathcal{L} = \{\pm 1\}$?	PROVED / REFUTED
$\mathcal{L}_Z$ central-product quotient identified?	PROVED / PARTIAL / REFUTED
Normalized full carrier $\cong Spin^c(3) \cong U(2)$?	PROVED / PARTIAL / REFUTED / NOT APPLICABLE
Full carrier $\cong \mathbb{R}_{>0} \times Spin^c(3)$?	PROVED / PARTIAL / REFUTED
Half-odd/ $\mathbb{Z}$ torsor given a standard one-parameter-group interpretation?	PROVED / PARTIAL / OPEN
Any principal spin structure constructed?	remain NO unless separately proved
Disconnected/analytic branches	remain OPEN unless formally affected

Methodological principle 7 (Task XXI success criterion). The next task succeeds if it converts the remaining near-resolutions into exact commuting diagrams or exact counterexamples. It does not need to enlarge the physics and should not broaden the search again unless one of the explicit candidates fails.

25 Source ledger and comparison provenance

The standard imported mathematics used directly by Task 20 includes ordinary topology of compact/Hausdorff quotients, trigonometric identities, intermediate value results, and — in the isolated comparison module only — Mathlib’s quaternion algebra [13, 12]. The reconstruction layer imports no conventional quaternionic or spin target.

The formally derived Task 20 layer includes the compact unit carrier, core and full internal carriers, their closure laws, projection and kernels, visible quotient/homeomorphism, relative-factor uniqueness, torsor laws, reversal quotient, dense-selector reformulation, minimal zero obstruction, and continuous-choice no-go. The exact external identification $\mathcal{L} = \text{unit quaternions}$ is derived in the comparison-only module.

The classical consequences discussed in sections 12 and 13 are sourced externally. Unit quaternions, $SU(2)$ and the standard double cover of $SO(3)$ are classical [2, 5, 14, 3]. The definition and low-dimensional identification of $Spin^c(3)$ are standard [4, 8, 10, 9]. These external facts are used to design Task 21; they are not retroactively inserted into Task 20’s formal claims.

26 Conclusion

Part XIX reached a level at which further progress required understanding how several independently reconstructed objects were related. Task 20 provides the first decisive resolution of that explanatory bottleneck.

The visible side is no longer an unnamed redundancy: its exact quotient exists, has the correct quotient topology, is homeomorphic to the visible image, and carries a closed group law. The internal side is no longer an unnamed “spin-like” carrier: its core is exactly the unit-quaternion group. The core projection has kernel $\{\pm 1\}$, while the larger central carrier has a genuinely continuous kernel equal to its centre. Equal visible lifts differ by unique central factors. The local exact rates form an exact $\mathbb{Z}$ -torsor rather than merely resembling one. The reversal quotient gives an exact selector interpretation of maximal admissible sets. And the global obstruction is reduced to its smallest form: a continuous reversal-odd rate must vanish, equivalently a quotient-compatible internal choice can be made set-theoretically but not continuously.

These results clear up the almost-resolutions of Part XIX. The broad comparison with several established theories was not wrong; it was underanchored. Task 20 supplies the anchor: unit quaternions. Classical mathematics then places that core directly in the $SU(2)/Spin(3)$ environment and supplies a standard conjugation map to $SO(3)$ with the same two-element kernel. The project has not yet proved that its visible image and projection are exactly those classical objects, but the remaining statement is now a concrete isomorphism problem rather than a thematic resemblance.

The enlarged carrier creates a second equally concrete target. Its central plane has complex multiplication, its core is quaternionic, and its full kernel is continuous. This makes a central-product/ $Spin^c(3)$ comparison precise enough to test, while the Task-20 kernel theorem prevents the simpler $\mathbb{Z}_2$ double-cover language from swallowing the larger central sector. The project may therefore contain a classical double-cover core inside a larger classical central extension; Task 21 must decide this explicitly.

The Conservation of Difficulty has consequently contracted rather than branched again. The question “what sort of mathematics have we reconstructed?” has become a finite collection of sharp statements:

Is the visible group exactly $SO(3)$ under the quaternionic core identification?
 Is the core projection exactly the standard $Spin(3) \rightarrow SO(3)$ double cover?
 Is the enlarged central carrier a central product related to $Spin^c(3)$, or does a specific inherited feature obstruct that identification?

That is why the project can now resume its normal targeted mode. The explanatory detour was necessary because the objects had not yet been named by theorem. One of them now has been. The next task can again proceed by explicit construction, kernel calculation, inverse map, and counterexample, without either assuming a conventional answer or reopening an unrestricted search.

No physical spin observable, state space, fermionic statistics, quantum dynamics or field equation follows from Part XX. Even a successful group-level $Spin(3)$ identification would not by itself construct a spin structure on a bundle or a physical spin degree of freedom. Those remain later questions, to be considered only after the present group and extension identifications are closed.

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